

ABBREVIATED CURRICULUM VITAE

Alejandro Guerrero-López · Updated 01 October 2026

PART A. PERSONAL AND PROFESSIONAL INFORMATION

Name: Alejandro Guerrero-López

Institution: Institute of Medical Microbiology, University of Zurich, Switzerland

Position: Marie Skłodowska-Curie Postdoctoral Fellow (since July 2026)

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Full CV: <https://aguerrerolopez.github.io/cv/>

A.1. Academic qualifications

2023 - PhD in Bayesian Machine Learning applied to Personalised Medicine, Universidad Carlos III de Madrid (UC3M). Thesis: Clinical microbiology with multi-view deep probabilistic models.

2020 - MSc in Information Health Engineering, UC3M.

2019 - BSc in Telematic Engineering, Universitat de les Illes Balears.

A.2. Research and academic appointments

July 2025-present - Postdoctoral researcher, University of Zurich; MSCA Fellow since July 2026, in Prof. Adrian Egli's group.

2024-2025 - Visiting professor in Data Science, UC3M.

2023-2024 - Postdoctoral researcher, Bioengineering and Optoelectronics Group, Universidad Politécnica de Madrid.

2020-2023 - Doctoral researcher, UC3M and Gregorio Marañón Health Research Institute.

2019-2020 - Research assistant, UC3M; 2017-2018 - Research assistant, Universitat de les Illes Balears.

PART B. CURRICULUM SUMMARY

My research focuses on applied machine learning for healthcare, with particular emphasis on clinical microbiology, Bayesian modelling and multimodal biomedical data. I develop methods for antimicrobial resistance prediction, bacterial identification and strain typing using routine MALDI-TOF mass spectra, and work with clinical collaborators to translate these methods into usable research tools.

My doctoral work connected probabilistic machine learning with hospital microbiology. Subsequent research at Universidad Politécnica de Madrid extended this approach to Parkinsonian speech, medical imaging and open biomedical datasets. At the University of Zurich, my MSCA project OUTBRAID addresses AI-driven outbreak detection using multimodal data.

My experience combines interdisciplinary research, open science, university teaching, ongoing doctoral co-supervision and peer review in machine learning and biomedical journals. These activities support the critical assessment of research methodology, validation, reproducibility and clinical relevance.

Research record: 16 published journal articles, 3 accepted journal articles and 5 preprints listed in the full CV. Google Scholar: 263 citations; h-index 9 (31 August 2026).

PART C. SELECTED RELEVANT MERITS

C.1. Selected publications

Selection spanning clinical microbiology, probabilistic machine learning and biomedical data. Accepted papers are explicitly identified.

1. Giacobbe, D. R., Ahmad, R., Akilli, F. M., Ascandari, A., Eyre, D. W., Gallardo-Pizarro, A., ... & Guerrero-López, A., on behalf of ESGAID and EAAS. (2026). How ready are we to use artificial intelligence in our fight against antimicrobial resistance? An ESGAID and EAAS perspective. *Expert Review of Anti-infective Therapy*, 24(1), 39-61. Last author. [Publication](#).
2. Guerrero-López, A., Sevilla-Salcedo, C., Candela, A., Hernández-García, M., Cercenado, E., Olmos, P. M., Cantón, R., Muñoz, P., Gómez-Verdejo, V., del Campo, R., & Rodríguez-Sánchez, B. (2023). Automatic antibiotic resistance prediction in *Klebsiella pneumoniae* based on MALDI-TOF mass spectra. *Engineering Applications of Artificial Intelligence*, 118, 105644. [Publication](#).
3. Candela, A.*, Guerrero-López, A.*, Mateos, M., Gómez-Asenjo, A., Arroyo, M. J., Hernández-García, M., del Campo, R., Cercenado, E., Cuénod, A., Méndez, G., Mancera, L., Caballero, J. D., Martínez-García, L., Gijón, D., Morosini, M. I., Ruiz-Garbajosa, P., Egli, A., Cantón, R., Muñoz, P., Rodríguez-Temporal, D., & Rodríguez-Sánchez, B. (2023). Automatic Discrimination of Species within the *Enterobacter cloacae* Complex Using Matrix-Assisted Laser Desorption Ionization-Time of Flight Mass Spectrometry and Supervised Algorithms. *Journal of Clinical Microbiology*, e01049-22. [Publication](#).
4. Blázquez-Sánchez, M.*, Guerrero-López, A.*, Candela, A., Belenguer-Llorens, A., Moreno, J. M., Sevilla-Salcedo, C., ... & Rodríguez-Sánchez, B. (2025). Automated web-based typing of *Clostridioides difficile* ribotypes via MALDI-TOF MS. *BMC Bioinformatics*, 26(1), 181. [Publication](#).
5. Schmidt-Santiago, L., Guerrero-López, A., Sevilla-Salcedo, C., Rodríguez-Temporal, D., Rodríguez-Sánchez, B., & Gómez-Verdejo, V. (2026). A systematic review of machine learning on clinical MALDI-TOF MS. *Briefings in Bioinformatics*, 27(3), bbag208. [Publication](#).
6. Sevilla-Salcedo, C., Guerrero-López, A., Olmos, P. M., & Gómez-Verdejo, V. (2022). Bayesian Sparse Factor Analysis with Kernelized Observations. *Neurocomputing*. [Publication](#).
7. Mendes-Laureano, J., Gómez-García, J. A., Guerrero-López, A., et al. (2024). NeuroVoz: a Castilian Spanish corpus of parkinsonian speech. *Scientific Data*, 11, 1367. [Publication](#).
8. Guerrero-López, A., Arias-Londoño, J. D., Shattuck-Hufnagel, S., & Godino-Llorente, J. I. (2026). A model for the automatic phonemic grouping of Parkinsonian speech. *Engineering Applications of Artificial Intelligence*. Accepted for publication.
9. Santiago, L. S., López-Mareca, I., Blázquez-Sánchez, M., Moreno, J. M., Sevilla-Salcedo, C., Gómez-Verdejo, V., Rodríguez-Sánchez, B., Rodríguez-Temporal, D., & Guerrero-López, A. (2026). MALDI-TOF MS spectra database from a Spanish University General Hospital (2018-2024). *Scientific Data*. Accepted for publication. Last author.

* Equal first authorship. Full publication list and supporting links:
<https://aguerrerolopez.github.io/publications/>.

PART C. SELECTED RELEVANT MERITS (CONTINUED)

C.2. Competitive funding

Marie Skłodowska-Curie Postdoctoral Fellowship, European Commission, 1 July 2026-30 June 2028. OUTBRAID - OUTBreak AI-driven Detection: Enhancing MALDI-TOF outbreak detection with multimodal AI. Host: University of Zurich; supervisor: Prof. Adrian Egli. Role: funded postdoctoral fellow.

C.3. Doctoral and thesis supervision

Ongoing doctoral co-supervision (4 projects; not completed doctorates):

- Luis Fuertes Pina - Development of machine learning models applied to FTIR and MALDI-TOF for improving microbiological diagnosis. Universidad Complutense de Madrid (UCM).
- Janis Rogenmonser - Advancing antimicrobial resistance prediction from whole-genome sequencing. Institute of Medical Microbiology, University of Zurich.
- Yukino Gütlin - Predicting antimicrobial resistance, virulence, and invasiveness from MALDI-TOF MS data. Institute of Medical Microbiology, University of Zurich.
- Eline Meijer - Boosting the resolution of MALDI-TOF mass spectra using transformer-based machine learning. Institute of Medical Microbiology, University of Zurich.

The full teaching record also lists 8 MSc and 3 BSc thesis supervision projects. Both ZHAW BSc projects were defended on 1 July 2026: MALDI-TOF/WGS data integration (5.5/6) and disc diffusion image analysis (4.5/6).

C.4. Teaching and scientific reviewing

Teaching in neural networks, deep learning, applied machine learning and AI for health at undergraduate and master's level. Nine official teaching evaluations: mean 4.83/5; multiple recognition letters from UC3M for teaching performance.

Co-coordinator with Alejandro Lancho Serrano of UC3M teaching innovation project No. 25 (2024-2025), integrating neural networks and generative AI in a real data engineering competition.

Peer review: Engineering Applications of Artificial Intelligence, Clinical Microbiology and Infection, IEEE Journal of Biomedical and Health Informatics, mSystems, and AISTATS.

C.5. Selected congress presentations and training

ESCMID Global 2026, Munich - Democratising MALDI-TOF AMR prediction: from Transformers to simple Neural Networks without accuracy loss, reducing energy costs via knowledge distillation. Oral presentation; presenting author. Abstract 2968/O0970 (21 April 2026).

ESCMID Global 2026, Munich - Digital AST Annotator: standardising Kirby-Bauer reading with an open-source app. Oral and ePoster presentation of one contribution; presenting author. Abstract 3082/E0842 (20 April 2026).

Instructor, ESCMID workshop on Artificial Intelligence and Machine Learning in Medical Microbiology Diagnostics, University of Zurich, 2-5 June 2025. Model-selection teaching materials openly available: [doi:10.5281/zenodo.15564228](https://doi.org/10.5281/zenodo.15564228).